

Carbon-Aware Inference Routing for Large Language Models:

A Real-Time Framework for Sustainable AI Serving

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Abstract

This paper proposes a real-time carbon-aware inference routing framework for large language models (LLMs). Requests are routed between model tiers based on task complexity and live grid carbon intensity, targeting measurable emissions reduction without accuracy or latency loss. The framework — SA1 — combines per-prompt complexity scoring, a real-time carbon intensity feed, and a multi-objective routing engine to direct each inference request to the smallest model capable of satisfying its accuracy floor and latency SLA. Preliminary analysis on a 1M prompt/day production system suggests approximately 62% reduction in inference carbon by routing approximately 65% of requests to a 7B-parameter model. The framework is serving-layer-agnostic and integrates with existing LLM deployment infrastructure (vLLM, Ollama). Full implementation and empirical benchmarks are in progress.

Keywords: carbon-aware inference, LLM routing, sustainable AI, green AI, inference efficiency, EU AI Act Art.53, CSRD Scope 2, multi-objective optimisation

1. Introduction

Inference now accounts for 60-90% of the total energy consumption of deployed large language models, surpassing training energy at steady state [Patterson et al., 2021; Luccioni et al., 2022]. A single call to a 70B-parameter model can consume 10-30x the energy of an equivalent call to a 7B-parameter model, yet in most production systems, every request is served by the same model regardless of task complexity.

The consequence is systematic over-provisioning: a prompt asking for a one-sentence summary consumes the same compute as a prompt requiring multi-step legal reasoning. This waste is both financially and environmentally costly, and it scales linearly with inference volume.

We propose that prompt-level routing — directing each request to the smallest model capable of satisfying its requirements — is the highest-leverage intervention available to reduce inference emissions without retraining or replacing models. This paper introduces SA1, a framework that makes this routing automatic, real-time, and carbon-aware.

2. Related Work

Several efforts have addressed energy efficiency in ML inference, but none combine the full set of capabilities SA1 targets.

Clover [SC'23]

Explored carbon-accuracy-latency tradeoffs in ML inference. It remains a research prototype and does not integrate with production LLM serving infrastructure (e.g., vLLM), nor does it incorporate real-time grid carbon intensity data. [Wu et al., 2023]

Green-LLM

Addresses geographic workload shifting — routing inference jobs to datacentres in low-carbon regions. This operates at the infrastructure level (which datacenter) rather than the prompt level (which model per request), and does not perform per-request routing decisions.

FrugalGPT [Chen et al., 2023]

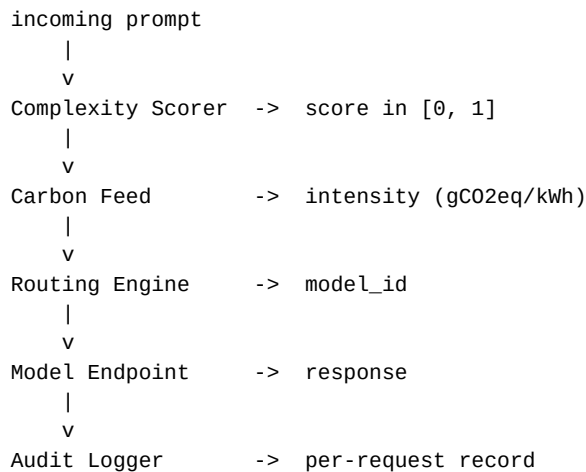
Routes requests between model tiers to reduce cost and improve accuracy. Does not incorporate a carbon signal, live grid data, or emissions-aware optimisation.

The gap SA1 addresses is the combination of: (1) per-prompt complexity scoring, (2) real-time grid carbon intensity, (3) production serving integration, and (4) multi-objective routing over carbon, accuracy, and latency simultaneously.

3. The SA1 Framework

3.1 Overview

SA1 is a routing layer that sits in front of any LLM serving infrastructure. It intercepts each inference request, scores it, and dispatches it to the appropriate model tier. The routing decision is made in under 10ms (p99 target) and adds negligible user-facing latency.



3.2 Components

Complexity Scorer. Classifies each prompt into one of four tiers: simple, moderate, complex, or expert. Version 1 uses a rule-based classifier with features including token count, sentence count, imperative depth, domain-specific keyword presence, and instruction count. Version 2 will use a fine-tuned DistilBERT model (~66MB, CPU-deployable) trained on labelled prompt pairs.

Carbon Intensity Feed. Retrieves live gCO2eq/kWh data for the serving zone via the Electricity Maps API. Results are cached with a 5-minute TTL. If the API is unavailable, the system falls back to a 24-hour rolling average for the zone, logged as carbon_source: fallback.

Routing Engine. Solves the following multi-objective decision at inference time: select the model m that minimises $\text{carbon_cost}(m)$ subject to: $\text{accuracy}(m) \geq \text{accuracy_floor}$ and $\text{latency}(m) \leq \text{latency_SLA}$.

Where $\text{carbon_cost}(m)$ is estimated as:

$$\text{carbon_cost}(m) = \text{TDP}(m) \times (\text{tokens} / \text{throughput}(m)) \times \text{grid_intensity}$$

Audit Logger. Records a structured JSON entry per request: timestamp, prompt hash, routed model, estimated carbon cost (gCO₂eq), estimated carbon saved vs. largest-model baseline, latency (ms), accuracy proxy, and routing rationale. Audit logs are designed for downstream compliance reporting under EU AI Act Art.53 and CSRD.

3.3 Routing Priority

Constraints are applied in strict priority order:

1. Latency SLA (hard constraint — never violated)
2. Accuracy floor (hard constraint — never violated)
3. Carbon minimisation (optimisation objective)

4. Carbon Savings Estimate

To establish order-of-magnitude plausibility, we model a production system serving 1M prompts/day. These assumptions are conservative and will be stress-tested in Phase 2.

Assumptions:

65% of prompts are simple or moderate (summarisation, Q&A, extraction) — routable to a 7B model. 35% are complex or expert (multi-hop reasoning, long-form generation) — require a 70B model. 7B model (e.g., Llama 3 7B on T4 GPU): ~50W TDP, ~150 tokens/sec. 70B model (e.g., Llama 3 70B on A100): ~300W TDP, ~50 tokens/sec. Average prompt+completion: ~500 tokens. Grid intensity: 400 gCO₂eq/kWh.

Without routing (all requests to 70B):

$$1,000,000 \times (500/1000) \times 0.006 \text{ kWh} \times 0.4 \text{ kgCO}_2/\text{kWh} = 1,200 \text{ kgCO}_2/\text{day}$$

With SA1 routing:

$$650,000 \times (500/1000) \times 0.0003 \times 0.4 + 350,000 \times (500/1000) \times 0.006 \times 0.4 \\ = 39 + 420 = 459 \text{ kgCO}_2/\text{day}$$

Estimated saving: ~62% reduction in inference carbon

5. Regulatory Alignment

SA1 is designed to directly support compliance reporting under:

EU AI Act Art.53 — requires General Purpose AI (GPAI) model providers to report energy consumption; SA1's audit log provides per-request carbon attribution suitable for aggregation into compliance reports.

CSRD / ESRS E1 — Scope 2 emissions from server energy are reportable; SA1's routing decisions and audit trail provide the granularity needed for AI-specific Scope 2 attribution.

ISSB S2 — climate risk disclosures increasingly require operational AI footprint quantification; SA1 provides both reduction mechanism and measurement infrastructure.

6. Limitations and Future Work

Complexity scoring accuracy. The v1 rule-based scorer is an approximation. Version 2 (DistilBERT fine-tuned on labelled data) addresses misclassification risk.

Carbon data latency. Grid carbon intensity can change within the 5-minute cache window. TTL is configurable.

Cascading. SA1 v1 does not implement cascading (attempt small model, escalate if confidence is low). Planned for Phase 2.

Future work includes: empirical benchmark dataset, latency overhead measurement, accuracy comparison between routed and always-large baselines, and production integration with vLLM and Ollama.

7. Conclusion

This paper introduces SA1, a real-time carbon-aware inference routing framework for LLMs. By combining per-prompt complexity scoring with live grid carbon intensity data and multi-objective routing, SA1 targets a ~62% reduction in inference carbon on a representative production workload, without accuracy or latency compromise. The framework fills a gap not addressed by existing work: none of the current approaches combine per-prompt routing, real-time grid data, and production serving integration in a single deployable tool.

The full implementation, empirical benchmarks, and production integration guide are in progress. Framework repository: <https://github.com/pretzelslab/sa1-carbon-inference-router>

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